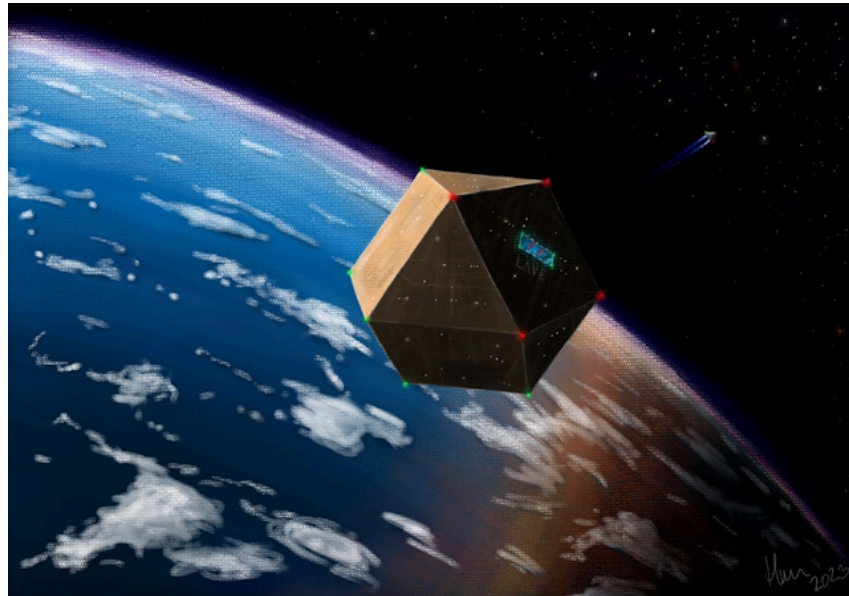


Elite: Unbound

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Leaving Lave by Ataribaby 2023

The Beginning

It all began around 1991, when, as a teenager, I discovered a program with the mysterious name **E L I T E** on my eight-bit **Didaktik M**, a locally produced clone of the **Sinclair ZX Spectrum 48K**. The program came on a cassette borrowed from someone. Being a fan of every kind of 3D simulator—most of which used see-through wireframe graphics at the time—Elite immediately captivated me with its 3D models, whose hidden edges were removed from view.

Over time, curiosity began to reveal the game's true depth. This was not merely a simple 3D shooter in the style of Starion or Starfox. There was something more: a complete simulation of life as a space trader, smuggler, bounty hunter, pirate, miner or interstellar playboy. Whatever you wanted to be, you could become.

The depth with which the game simulated space, spaceflight, combat, economics and trading drew me in completely. That summer, going outside held very little appeal. After a year of playing, my alter ego finally attained the status --- ELITE ---. Fortunately, a detailed guide appeared around that time in the games magazine FIFO, finally answering the questions that had remained unresolved without the original manual.

Back then, rumours of **Rock Hermits**, generation ships and other mysteries sent me searching across the galaxy. The Rock Hermits eventually turned up; the generation ships never did. A neighbour once claimed to have encountered a ship that remained invisible most of the time and could be seen only on the scanner. According to him, destroying it released a cargo canister, and scooping it revealed something called a Cloaking Device.

Unfortunately, he never reached a station to save his progress. Pirates destroyed his ship on the way, before he could discover how to use the device. At the time, not a word of the story seemed believable. Today, of course, I know he was telling the truth. The ZX Spectrum version contains a special Asp Mk II fitted with exactly such a device. Once the ship has been destroyed and its canister scooped, pressing Y activates the Cloaking Device.

Time passed. Secondary school began, the ZX Spectrum clone gave way to an Amiga 1200, and Elite cast its spell once more—this time as **Frontier: Elite II**. But that is another story...

That period also brought a **C64** and a first encounter with its version of Elite, which is very close to the original BBC Micro release. The ZX Spectrum version differs in several respects and works somewhat differently. This is because its developers received only the procedural universe-generation code from the original creators. Everything else was recreated by playing the original game on the BBC Micro.

That history is also why *Elite: Unbound* aims to echo the aesthetics of the ZX Spectrum version. The Spectrum edition remains my favourite for one simple reason: it was my first.

For me, the original Elite defined the open-world sandbox built around procedural generation. The game has no ending and no fixed objective beyond attaining ELITE status. Everything else belongs to the player, including the story they choose to create. This is the kind of game I still love and seek out today.

And now it is 2026, and technology has advanced beyond belief. What once existed only in my imagination—as a simple circle representing a planet grew larger and a space station drifted above its horizon—has become a reality that modern **Elite: Dangerous** renders almost perfectly.

Another dream has come true as well: creating the version of *Elite* I always wanted and being able to offer it here. Hopefully, you will enjoy it too.



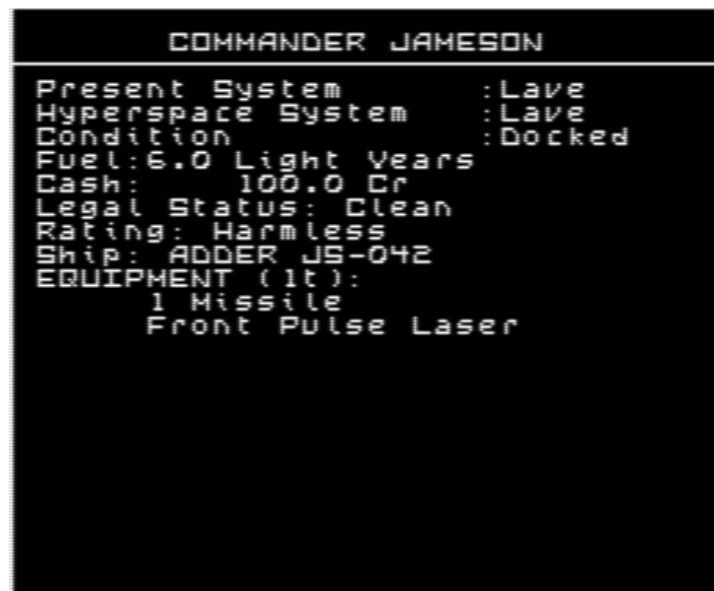
What is *Elite: Unbound*?

Elite: Unbound is a source-code modification of the original Commodore 64 version of *Elite*. It is built on the complete and extensively documented BBC Micro and Commodore 64 source code made available by **Mark Moxon**. Without his work, this project would not have been possible.



Title screen

The modification runs on a standard Commodore 64 with either a cassette or disk drive; no additional hardware is required. It is distributed as D64 and TAP images for use in a C64 emulator, or for transfer to physical media and use on a real C64.



Starting Commander

The main goal of *Elite: Unbound* is to give player greater freedom, more meaningful choices and a stronger sense of immersion. Ships other than the Cobra Mk III can now be purchased and flown. Each has its own agility, speed, hyperspace range, shield strength and hull durability. Cargo and equipment capacity also varies from ship to ship: every piece of internal equipment has a mass and reduces the space available for cargo, making each loadout a deliberate compromise. Ships also differ in the number of laser mounts and guided-missile pylons they provide.



Buying a ship

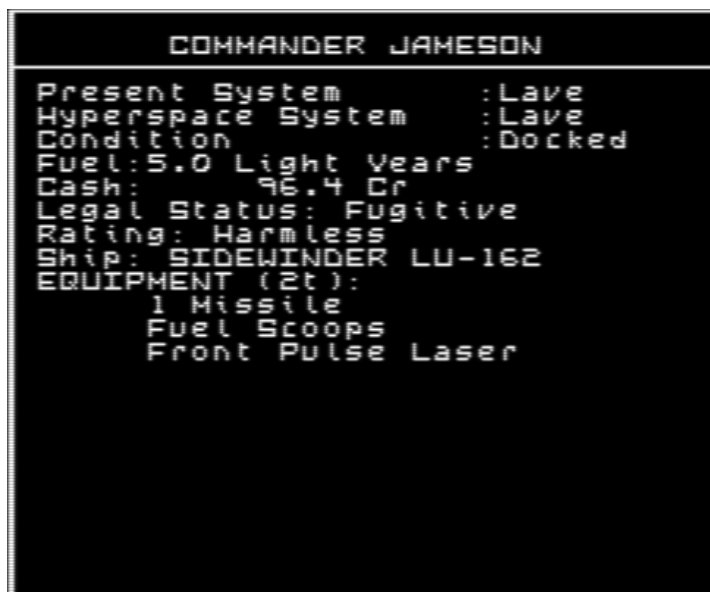
This also means that player can sell installed equipment.



Selling installed equipment

The player now occupies a more balanced position within the game world. The Energy Bomb—an advantage unavailable to AI ships—has been removed and replaced by an I.F.F. transponder and interrogator for identifying other vessels. Guided missiles have also been reworked, and a hit no longer automatically destroys its target.

A new career can begin in the modest Adder, which is also the default starting ship, or continue in the capable all-round Cobra Mk III. Another option is the Sidewinder: small and fragile, but fast and agile, which explains its popularity among pirates. Its speed may be the only way out of a fight. The choice of ship—and the kind of story it supports—now belongs to the player.



Pirate's Sidewinder

Stations will not launch other vessels while either the player's ship or an AI ship is blocking the entrance corridor, reducing the risk of collisions. Police ships are naturally exempt from this restriction.

Basic radio communications have been introduced. A station will warn ships that remain in its approach corridor longer than necessary.



A Coriolis station warns the player about loitering in the approach corridor

AI ship spawning has been reworked so that it is no longer immediately obvious whether a new contact is a trader or a pirate. Space debris—including cargo canisters, wreckage and asteroids—no longer follows the player during a local jump; it disappears instead.

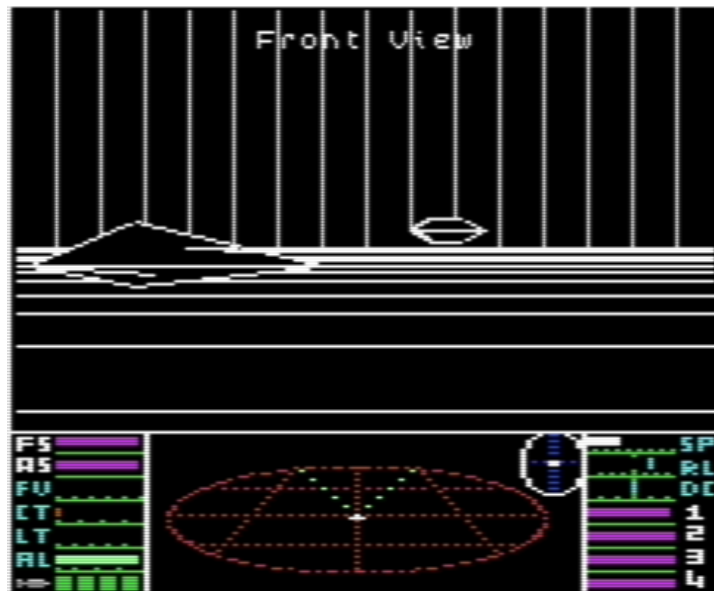
Cargo can now be jettisoned in flight, allowing pirates to decide which spoils are worth keeping. Dumping cargo into open space is, of course, prohibited near most stations.

Every ship, including the player's, now carries a unique registration ID in the format AA-123. It is used for communication and identification, and can be displayed by the I.F.F. system. Pirates and other questionable operators can conceal their registration, and the player may eventually discover how to do the same. The choice comes with both advantages and consequences.



The I.F.F. system returns the targeted ship's registration ID

The post-docking hangar screen familiar from the BBC Micro version has also been restored.



Hangar view after docking

Bounty information now reports the actual bounty value rather than a combined total.

Visual changes include a new white-themed cockpit, the font from the ZX Spectrum version, and single-beam laser fire. Ships are represented as having single-barrel lasers—as suggested by the weapon mounted on the nose of the Cobra Mk III—rather than twin barrels, and the player's laser now matches the fire used by AI ships. Every laser type has a crosshair, while the revised docking-tunnel effect more closely matches the shape of a station's docking aperture.

Technical improvements include a frame-rate limiter set to 12.5 FPS on PAL systems, a revised keyboard and joystick handler that allows both input methods to work simultaneously, and faster circle drawing, particularly when rendering planets.

Flicker-free foundation

This modification includes the backported flicker-free ship drawing and planet fixes from Mark Moxon sources.

Such an extensive expansion inevitably comes at a cost. Like every home 8-bit computer of its era, the C64 has very little memory to spare. To make room for the new features, all music was removed and the instrument panel was compressed using RLE compression.

Story continues...

Existing save games from Commodore 64 *Elite* remain compatible. When loaded, they are automatically converted to the *Elite: Unbound* format.

Features

Highlights of the main *Elite: Unbound* features.

- New games begin in a puny Adder
- Alternative starting positions supplied as save games in C64 TAP format
- Ability to buy and fly multiple ship types
- Ability to sell installed ship equipment
- An I.F.F. Unit replacing the Energy Bomb
- Unique registration IDs for all AI-controlled ships and the player's ship
- Realistic missile damage against AI ships, with larger vessels no longer destroyed by a single missile
- Scaled collision damage according to relative ship size
- In-flight cargo jettison
- A richer pirate experience
- Randomized AI ship spawn positions

- Expanded Status and Inventory screens
- Bounty messages showing only the bounty awarded
- A new white-themed cockpit
- Protected station approach corridors
- Revised station launch and docking procedures that prevent collisions with AI ships leaving the station
- AI radio chatter for traders, police, bounty hunters, pirates and stations
- Restored docking hangar scene from BBC Micro Elite
- Concentric rectangular docking-tunnel frames replacing circular rings to better match the station's docking aperture
- Faster circle rendering, especially for planets
- Improved keyboard and joystick input handling
- A frame-rate limiter
- Disk and tape versions
- Still runs on a stock Commodore 64 without additional hardware
- Compatibility with original Commodore 64 Elite save games

Right on, Commanders!

Instruction Manual

RTFM!

If you are completely new to the world of the original Elite, be sure to read the [original Elite C64 manual](#). The manual below covers only the additions introduced by *Elite: Unbound*.

This PDF is my own scan of the manual included with my original copy of Elite for the C64.

New Keyboard Commands

While in flight:

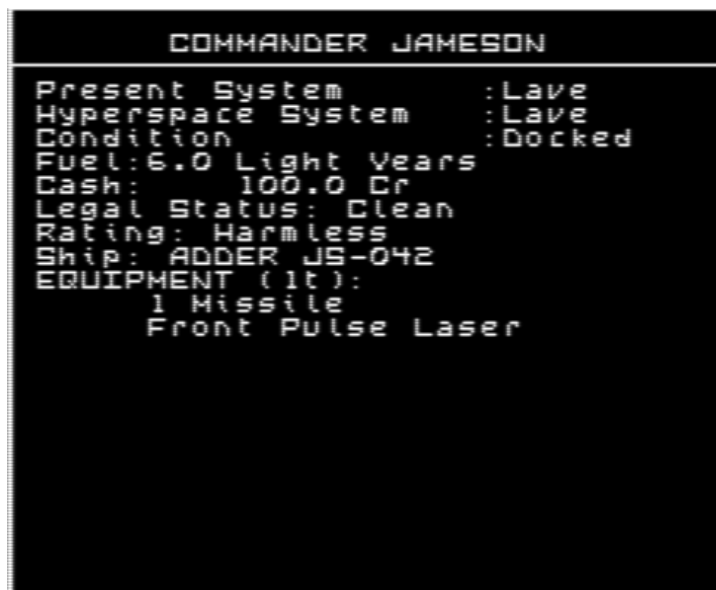
- I — Activates the I.F.F. interrogator.
- Ctrl + 2 — Opens the Cargo Jettison screen.

While docked:

- Ctrl + 2 — Opens the Sell Equip screen.
- Ctrl + 3 — Opens the Buy Ship screen.

Status Screen

The Status screen has been expanded to display additional information.



Ship: ADDER JS-042

Shows the type of your current ship and its registration ID.

EQUIPMENT (1t)

Shows the total weight of all installed equipment, excluding guided missiles. Missiles occupy dedicated slots in the ship's hull and therefore do not take up space in the cargo hold. The total weight of installed equipment is deducted from the ship's maximum cargo capacity. Consequently, the more equipment you install, the less room remains for cargo.

Inventory Screen

The Inventory screen has been expanded to display additional information.

INVENTORY (0t/8t)		
Fuel:	6.0	Light Years
Cash:	1.2	Cr
Food		1t
Textiles		2t
Radioactives		3t
Liquor/Wines		1t

INVENTORY (0t/8t)

Shows the remaining cargo capacity followed by the ship's total cargo capacity. The weight of installed equipment is deducted from the total, and the remainder is available for cargo. In this example, the Adder's unmodified cargo hold has a total capacity of 8t. The installed Front Pulse Laser accounts for 1t, as indicated by **EQUIPMENT (1t)** on the Status screen, while cargo occupies the remaining 7t. This leaves 0t of free capacity.

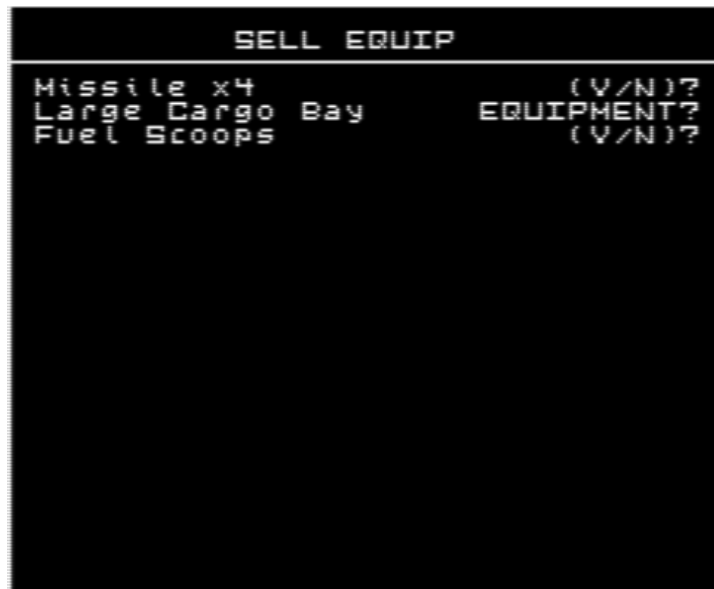
Installing a Large Cargo Bay increases the total cargo capacity by an amount that depends on the ship type. This increase is automatically included in the displayed total.

Sell Equip Screen

Installed ship equipment can now be sold. Press Ctr1 + 2 while docked to open the Sell Equip screen. Equipment is bought back for 50% of its purchase price.

SELL EQUIP	
Missile x1	(Y/N)?
Front Pulse Laser	(Y/N)?

Each installed item is offered for sale in turn with a **(Y/N)?** prompt. The missile entry displays the total number fitted, but each confirmation sells only one missile. To sell additional missiles, open the Sell Equip screen again.



A Large Cargo Bay can be sold only if the ship's current load will fit within the standard cargo hold after the extension is removed. Otherwise, **EQUIPMENT?** appears when the combined weight of cargo and installed equipment exceeds the standard capacity, while **CARGO?** appears when the cargo alone will not fit.

Replacing one laser with another on the Equip Ship screen follows a different, older rule: the fitted laser is credited at 100% of its purchase price. Selling a laser directly through the Sell Equip screen returns only 50%.

Buy Ship Screen

New ships can now be purchased. Press Ctr1 + 3 while docked to open the Buy Ship screen.

The screen lists the ships currently offered for sale and their purchase prices. Availability depends on the system's economy, with additional black-market ships offered in Anarchy systems.



Before buying a new ship, you must sell all ordinary installed equipment, as it cannot be transferred to the new hull, along with any excess missiles that the new ship cannot carry. The Naval Energy Unit is exempt and transfers to the new ship.

When you purchase a new ship, your existing cargo is retained. It is then checked to ensure that it fits within the new ship's **standard cargo hold**. A transferred Naval Energy Unit also counts as 1t.

A purchase may be rejected with one of the following messages: **CASH?**, **EQUIPMENT?**, **CARGO?**, or **MISSILES?**.

Equip Ship Screen

The **I.F.F. Unit** has been added to this screen and becomes available for purchase at Tech Level 7 and above.

EQUIP SHIP		
1	Fuel	0.0
2	Missile	25.0
3	Large Cargo Bay	400.0
4	E.C.M. System	400.0
5	Extra Pulse Lasers	400.0
6	Extra Beam Lasers	1000.0
7	Fuel Scoops	375.0
8	Escape Pod	200.0
9	I.F.F. Unit	1250.0
10	Extra Energy Unit	700.0
11	Docking Computers	800.0
12	Galactic Hyperspace	3000.0
13	Military Laser	6000.0
14	Mining Laser	800.0
Item?		

The only other change is that equipment prices now vary according to the type of ship you are flying.

In particularly disreputable systems, you may even find some less-than-legal options here.

In-Flight Cargo Jettison

A single 1t cargo canister can now be jettisoned while in flight. Press Ctrl + 2 to open the Jettison screen.

JETTISON		
Food	1t	(Y/N)?
Textiles	2t	(Y/N)?
Radioactives	1t	(Y/N)?
Liquor/Wines	2t	(Y/N)?
Luxuries	1t	(Y/N)?

Eligible cargo types are presented in turn with a **(Y/N)?** prompt. Any commodity carried in one-tonne canisters can be selected, except **Alien Items**. Confirming a choice ejects one 1t canister. The canister retains the selected commodity, so recovering it with Fuel Scoops returns the same type of cargo.

A successful jettison is confirmed by a beep, after which the **INVENTORY** screen is displayed.

WARNING! Jettisoning cargo within the Safety Zone of most stations is illegal and can adversely affect your legal status.

I.F.F. Transponder and Interrogator Operations

This new device provides early warning of hostile vessels and identifies other ships.



The I.F.F. Unit is fully integrated with the guided-missile targeting system and can also operate independently when no guided missiles are aboard.

Use **I** to activate the I.F.F. Unit. When guided missiles are aboard, the **I** key performs the same targeting function as the missile-targeting key **T**, and targeting proceeds in exactly the same way. Once a target is acquired, its registration ID is displayed if it is being transmitted; otherwise, **??-???** appears.

If no guided missiles remain aboard, pressing **I** activates the interrogator. A beep signals that it is ready to acquire a target, much like missile targeting. Once a target is acquired, its registration ID is displayed and the interrogator switches off. Press **U** to switch it off manually before a target is acquired.

Transponder operation is automatic and requires no interaction. Your ship's registration ID is broadcast continuously, as required by GalCop for every spaceborne vessel.

Hostile contacts are also identified on the scanner.



These contacts have a **T**-shaped blip head instead of the rotated **L** shape used for other contacts.

Bounty Display

After a target is destroyed, the upgraded bounty display shows the bounty awarded for that target rather than your total credit balance after the reward has been added.



Station Docking Procedure

Stations now have a protected approach corridor extending outward from the docking entrance.



Any ship that loiters in this corridor receives a loitering warning from the station. No ship is cleared for launch while the corridor remains occupied.

Radio Communication



If you receive a communication from another ship or station while viewing a screen other than the cockpit, a short beep alerts you to the new message. When you return to the cockpit, the last received message is displayed.

Credits & Foundations

Elite was written by Ian Bell and David Braben. Commodore 64 Elite is copyright © D. Braben and I. Bell, 1985.

Alongside my own modifications, *Elite: Unbound* builds on Mark Moxon's documented Commodore 64 source code and flicker-free work. The player-ship concept and data model are inspired by Angus Duggan's Elite-A, with additional ideas drawn from Uz's Elite 128 for the C64. *Elite: Unbound* is maintained by Ataribaby.

- https://elite.bbcelite.com/playing_elite/
- <https://github.com/markmoxon/elite-source-code-commodore-64>
- <https://github.com/markmoxon/elite-a-source-code-bbc-micro>
- <https://csdb.dk/release/?id=176103>
- <http://www.elitehomepage.org/>

None of my work would be possible without the enormous effort of everyone mentioned above. I am merely a LEGO builder here.

This is an educational, non-profit source-code project created in appreciation of one of the defining games of the 8-bit era.